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Josephson-Effect Far-Infrared Detector

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Superconducting point contacts, bridges, and tunnel junctions are all capable of carrying a zero-voltage current; i.e., no voltage appears across the device until the current exceeds a finite critical value. The critical value of the zero-voltage current is changed when microwave radiation is incident on such a junction. Thus such a device can be used as a detector of radiation. Experiments have been carried out demonstrating that this detection mechanism extends well into the far-infrared, is highly sensitive, and has inherently high speed. Junctions were formed by pressing together the ends of two superconducting wires, one of which was flat, the other pointed. The spectral response of such point contact junctions was studied by using them as detectors in a far-infrared Fourier transform spectrometer in which they were irradiated by broadband, incoherent radiation. The response of Nb-Nb junctions was found to extend to frequencies above 40 cm^{-1} ($\lambda < 250\mu$), i.e., to about twice the superconducting energy gap. Measurements using a klystron source at 2.5 cm^{-1} (λ=4 mm) yielded a value of 5×10^{-12} W for the noise equivalent power in a one-cycle bandwidth and showed the junction detector could follow a pulsed signal which had a risetime of 10 nsec. Preliminary experiments using a monochromatic laser source at 32.2 cm-1 showed the appearance of constant-voltage steps in the voltage-current characteristic of the junction, as is well known at microwave frequencies. These experiments demonstrate the existence of the Josephson effect at frequencies up to and beyond the superconducting energy gap: and show that over this range of frequencies, Josephson junction detectors exhibit both high sensitivity and high speed when compared with other helium-temperature far-infrared detectors.

I. INTRODUCTION

The Josephson effect[1] refers to certain remarkable properties of current flow through a barrier or weak link between two superconductors. Direct current can be driven through such a structure (called a Josephson junction) without developing any voltage across the junction. The maximum amplitude of this zero-voltage current is a quasiperiodic function of an applied magnetic field.* When a dc voltage V is maintained across a junction, the Josephson current is oscillatory with a frequency proportional to the voltage $\nu = 2eV/h$, where e is the magnitude of the electronic charge and h is Planck's constant. A voltage of 1 mV corresponds to a frequency of 16.12 cm^{-1} (483.6 GHz). Radiation emitted by these alternating Josephson currents has been observed by a number of workers.[3-6] When a junction is exposed to an electromagnetic field, the dc voltage-current characteristic is modified. Thus the junction can be used as a detector of radiation. This effect was first used however to demonstrate the existence of the ac Josephson effect.[7,8] The alternating Josephson currents are frequency-modulated by rf voltages driven by the applied monochromatic radiation. Whenever one of the modulation sidebands occurs at zero frequency, a current step at constant voltage appears in the *V-I.* characteristic of the junction. This happens for voltages such that the frequency of the alternating Josephson current is equal to an integer n times the frequency f of the applied radiation, $V = nhf/2e$. The amplitude of the current in the nth step varies with the radiation power level as the Bessel function of order n, $J_n(2ev/hf)$, where v is the equivalent voltage amplitude of the applied radiation. One consequence of this frequency modulation is that there are many discrete frequencies simultaneously present in a Josephson junction. This has been verified in the radiation experiments referred to earlier,[3-8] in which the radiation "feeds back" on the junction, and in harmonic-generation experiments.[9]

In this paper, however, the detection aspects of the Josephson effect are emphasized, and, in particular, detection at frequencies up to and beyond the superconducting energy gap. Experiments are described which show the Josephson-effect detector to have a frequency response estending into the submillimeter region. In addition, our esperiments demonstrate the high sensitivity and inherently fast speed of response of such detectors. A preliminary report of our work has already appeared.10 Here complete details of the experi-

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[1] B. D. Josephson, Phys. Letters 1, 251 (1962); Rev. Mod. Phys. 36,216 1964) ; Advan. Phys. 14, 419 (1965).
[2] P. W. Anderson and J. M. Rowell, Phys. Rev. Letters 10,230 (1963); J. hf. Rowell, Phys. Rev. Letters 11, 200 (1963).
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[4] D. N. Langenberg, D. J. Scalapino, B. N. Taylor, and R. E. Eck, Phys. Rev. Letters 15, 294, 842E (1965).
[5] A. H. Dayem and C. C. Grimes, Appl. Phys. Letters 9, 47 (1966).
[6] J. E. Zimmerman, J. A. Cowen, and A. H. Silver, Appl. Phys. Letters 9, 353 (1966) ; A. H. Silver and J. E. Zimmerman, Appl. Phys. Letters 10, 142 (1967) ; J. E. Zimmerman and A. H. Silver, Php. Rev. Letters 19, 14 (1967).
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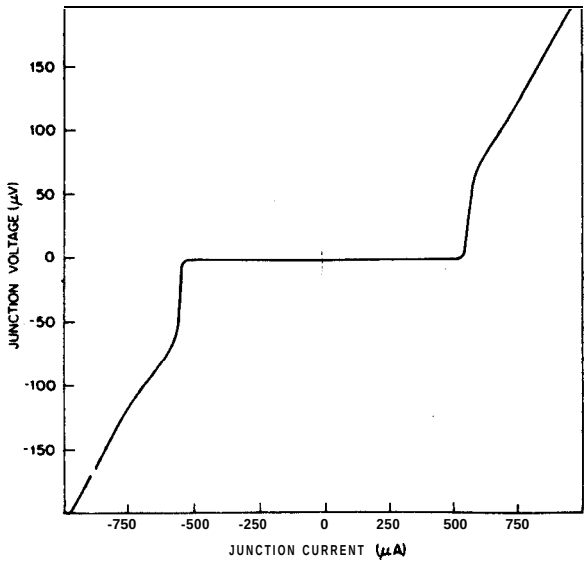


FIG. 1. Representative voltage-current characteristic for a Nb-Nb point-contact Josephson junction at 4.2°K.

ments and new results are presented. In addition, experiments are reported in which one junction, used as a detector, was affected by another physcially separated junction which was treated as an emitter of radiation.

II. JUNCTION PREPARATION AND CHARACTERISTICS

A number of different forms of Josephson junction have been used in studies of the Josephson effects. In our work point-contact junctions[11] were used because their low junction capacitance permits better coupling to radiation at the high frequencies where most of our experiments were done. In addition, the point-contact junctions are easy to fabricate and their V-I characteristics are readily adjusted.

The junctions were formed by pressing together the ends of two pieces of wire, one of which was flattened and the other sharpened by filing, by cutting with a razor blade, or by chemical etching. Thirty mil diam wire proved to be a convenient size. For junctions involving niobium (Nb) or tantalum (Ta), wires of those metals were used directly; for junctions with indium (In) or lead (Pb), copper wire tinned with In or Pb was used.

By varying the contact pressure, the V-I curve of a junction could be adjusted while the junction was immersed in liquid helium. The mechanism for varying the contact pressure involved the adjustment by a differential screw of a spring-loaded arm which pivoted about a bearing axis transverse to the axis of the point contacts. (An improved mechanism, also employing a differential screw, has been designed and tested by Contaldo.[12])

A representative V-I characteristic, here for a Nb-Nb point-contact junction at 4.2°K, is shown in Fig. 1. The presence of zero-voltage current is clearly indicated. The masimum amplitude of the zero-voltage current oscillated with applied magnetic field, but remained finite. This indicates the existence of the dc Josephson effect in such junctions, but differs somewhat from the behavior of thin-film tunnel junctions in which the zero-voltage current is actually reduced to zero at certain field values. Note that when voltage appears across the junction at the end of the zero-voltage region, it increases smoothly and continuously with increasing current. This too is in contrast to the case of thin-film tunnel junctions where at small but finite voltages the dc current carried by the junction falls to a small fraction of the zero-voltage current. The smooth transition from zero-voltage to finite voltage as shown in Fig. 1 is a characteristic of point-contact, or point-contact-like junctions and has also been observed by other workers.[5,9,11,13] (Recently, Werthamer and Shapiro have shown from analog computer studies of Josephson radiation effects[14] that such a smooth transition can occur for a junction coupled tightly to one or more electromagnetic resonators.)

Occasionally, particularly at higher contact pressures and consequently larger zero-voltage currents, the junction would switch discontinuously out of the zero-voltage region to a finite voltage part of the characteristic. Such switching appeared as a hysteresis region

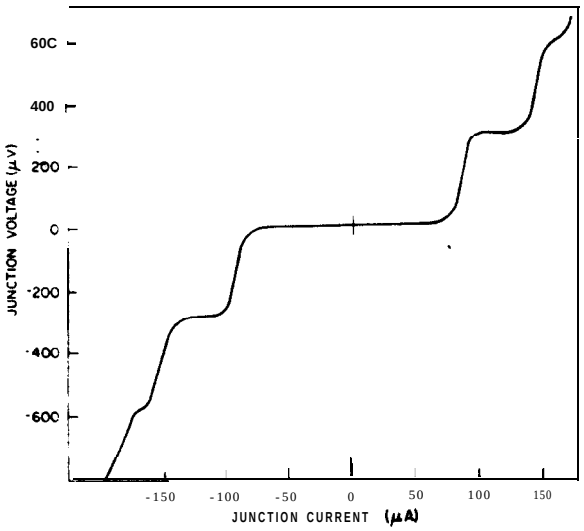


FIG. 2. V-I characteristic for an In-In point-contact Josephson junction at 1.3°K in the presence of about 10^{-9} W of radiation at 5 cm^{-1} (150 GHz).

[11] J. E. Zimmerman and A. H. Silver, Phys. Rev. 141, 367 (1966).

[12] A. Contaldo, Rev. Sci. Instr. 38, 1543 (1967).

[13] J. I. Pankove, Phys. Letters 21,406 (1966) ; Phys. Letters 22, 557 (1966).

[14] K. R. Werthamer and S. Shapiro, Phys. Rev. 164, 523 (1967).

on an oscilloscope display. Most of the data presented in this paper were taken for characteristics free of hysteresis regions. The exceptions were Pb-Pb junctions and a few other cases to be discussed later.

The existence of the ac Josephson effect in these adjustable point-contact junctions was verified by applying monochromatic radiation at about 2.5 cm^{-1} (75 GHz) of sufficient intensity to generate a number of constant-voltage steps, properly spaced in voltage, in the V-1 curve. It was possible with modest power to generate steps at voltages beyond the energy gap 2A of the superconductor used in the junction, i.e., at voltages higher than $V = 2\Delta/2e$. The current amplitude in these steps varied roughly in the usual Bessel function manner with rf power level. In a number of cases, additional steps were seen at voltages related to subharmonics of the applied frequency. These steps are characteristic of narrow metallic bridges between superconductor+ and their presence indicates the complex nature of point-contact junctions. The subharmonic steps tended, to appear, and the current in them to become appreciable, for junctions with zero-voltage currents larger than a few milliamperes. Since we usually worked with junctions whose zero-voltage current was less than 1 mA, we tended to avoid junctions with subharmonic steps. Nevertheless, in those cases where junctions with subharmonic structure were measured no drastic differences in detection behavior were noted.

A few experiments were done with monochromatic radiation at about 5 cm^{-1} (150 GHz) incident on a junction. A small amount of power at this frequency, estimated at a masimum of 10^{-8} W, was obtained by doubling from 2.5 cm^{-1} in a semiconductor harmonic generator. Figure 2 shows the V-I curve in an In-In point-contact junction esposed to the 5 cm^{-1} radiation. Even at this frequency which is higher than one-half the indium superconducting energy gap, constant-voltage steps characteristic of the ac Josephson effect are present. The presence of steps in the V-I curve at such low power levels demonstrates that point-contact Josephson junctions show high sensitivity as 5 cm^{-1} video detectors without optimizing the coupling of the junction to the radiation field.

III. FAR-INFRARED EXPERIMENTS

To study the frequency response of Josephson junctions, a series of experiments were done in which point-contact Josephson junctions, formed from a number of different superconductors, were used as detectors in a far-infrared Fourier-transform spectrometer. The design, principle of operation, and performance characteristics of the spectrometer have been reported by Richards. [16] In this type of spectrometer, broad-band radiation of very low power falls on the junction so no constant-voltage steps can be seen. The detector output is taken from the change in zero-voltage current produced by the radiation.

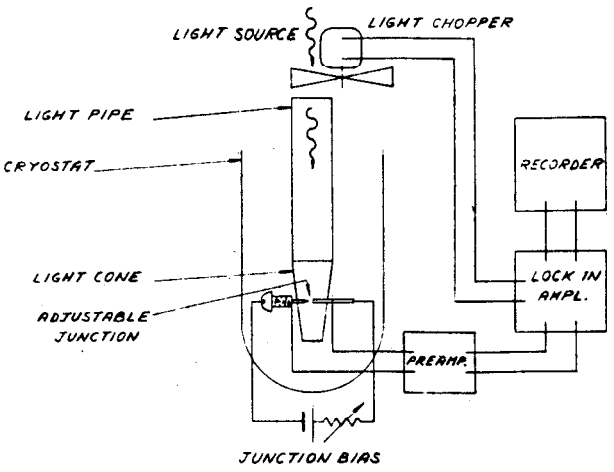


FIG. 3. Schematic of far-infrared detection experiment.

Broad-band, incoherent radiation from a mercury arc lamp illuminated a Michelson interferometer, one arm of which was adjustable. After the light was recombined, it was chopped at about 90 Hz and focused into a light pipe. As shown in Fig. 3, the light passed down the light pipe into the liquid helium Dewar and was focused by a cone onto the adjustable point-contact junction. The junction was mounted transverse to the cone axis and was immersed directly in the liquid helium. The mechanism used to adjust the junction characteristic has been described in Sec. II.

Leads were brought in to provide bias for the junction and to permit measurement of the V-I curve. To minimize pickup and the harmful effect of sparks or other electrical transients on the junction characteristic, the leads were twisted and shielded wherever possible both inside and outside the cryostat.

The radiation diminished the maximum amount of zero-voltage current that could flow through the junction. A voltage at the chopper frequency related to the change in zero-voltage current was developed across the junction and served as the detector output. Figure 4 shows schematically how this was done. The solid curve shows the junction characteristic in the absence of radiation. The bias current through the junction was held fixed at a point of high differential resistance. In the presence of radiation, the characteristic shifted to that indicated by the dashed line, so a voltage appeared across the junction at the chopper frequency. This voltage, suitably amplified and rectified in a lock-in amplifier, was plotted on a chart recorder as a function of the path difference in the interferometer. Depending on the junction impedance, either a low or high input impedance preamplifier was used. Although the signal voltage at the preamplifier input varied with impedance

[15] P. W. Anderson and A. H. Dayem, Phys. Rev. Letters 13,195 (1964); A. H. Dayem and J. J. Wiegand, Phys. Rev. 155, 119 (1967).

[16] P. L. Richards, J. Opt. Soc. Am. 54, 1474 (1964).

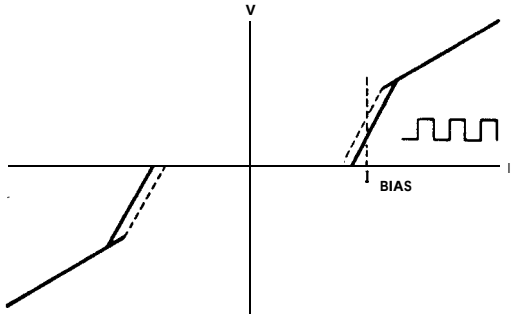


FIG. 4. Schematic V-I curve showing how applied radiation causes a change in zero-voltage current which is converted to a voltage output by constant-current external bias. Dashed curve is for radiation on; solid curve for radiation off.

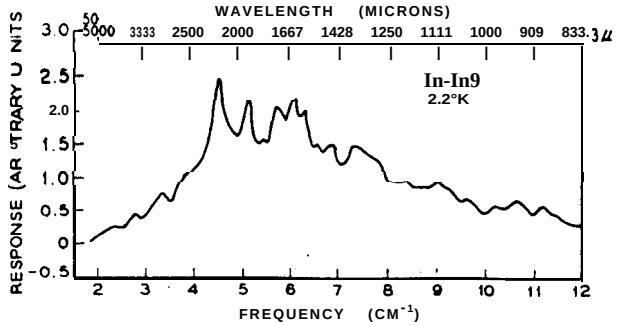


FIG. 6. Spectral response of an In-In point-contact Josephson junction at high resolution. This curve was obtained by computing the Fourier transform of an interferogram like that in Fig. 5. Structure is associated with resonances of the experimental mounting geometry.

from a few nanovolts to a few microvolts, it was possible to obtain good signal-to-noise ratios over the entire range of junction impedances tested, i.e., from a few tenths of an ohm to a few hundred ohms. The variability in signal-to-noise from run-to-run was somewhat less for high-impedance junctions; hence such junctions were used predominantly.

Figure 5 shows a representative interferogram recorded with an 0.3 sec time constant. The spectral response of the Josephson junction detector was obtained by computing the Fourier transform of the interferogram on a digital computer.[16]

The spectrum of an In-In point-contact junction is shown in Fig. 6. This spectrum has an instrumental resolution of 0.1 cm^{-1} and shows structure associated with the large number of weak standing-wave resonances of the conical mounting geometry. In other experiments, using a modified coaxial cavity, very pronounced structure was obtained associated with cavity resonances. The conical mounting eliminated most but not all such structure. As shown in Fig. 7, for the same In-In junction, low-resolution spectra (here, about 1 cm^{-1}) averaged over these geometric resonances. Generally it is such low resolution spectra that are shown in the balance of this paper.

Despite a certain amount of variability from junction-to-junction, the major features shown in Fig. 7 were common to all In-In junctions tested, provided their V-I curves were free of hysteresis regions. There are two main points to be noted about these data. First, there is a peak in the response located near to, but significantly below, the indium superconducting energy gap, which for this temperature lies at about 7.2 cm^{-1}. Second, the response is broad extending in frequency to about twice the energy gap. At lower temperatures the peak sharpened slightly.

These results for In-In junctions are suggestive of the predictions of Riedel[17] and Werthamer[18] who

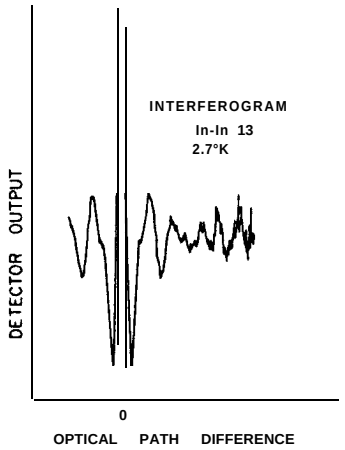


FIG. 5. Representative interferogram showing form of ran experimental data. This curve was obtained by plotting the detector output vs the optical path difference in a far-infrared Michelson interferometer through which broad-band radiation passed before reaching the detector. The system time constant was 0.3 sec.

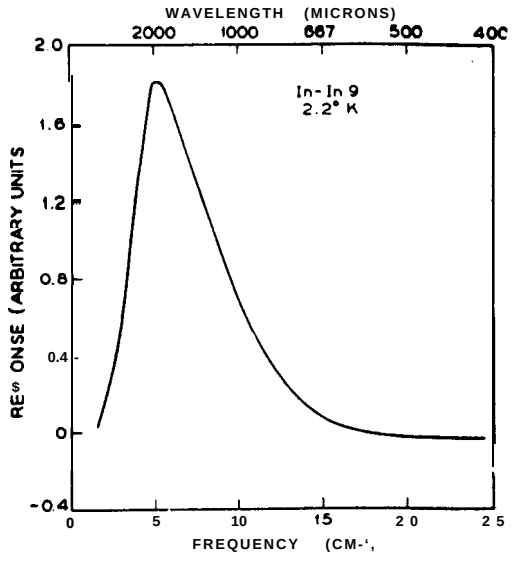


FIG. 7. Spectral response of the In-In junction of Fig. 6 at low resolution, which smooths out and eliminates structure associated with geometrical resonances.

[17] E. Riedel, Z. Saturforsch. 19a, 1634 (1964).
[18] N. R. Werthamer, Phys. Rev. 147, 255 (1966).

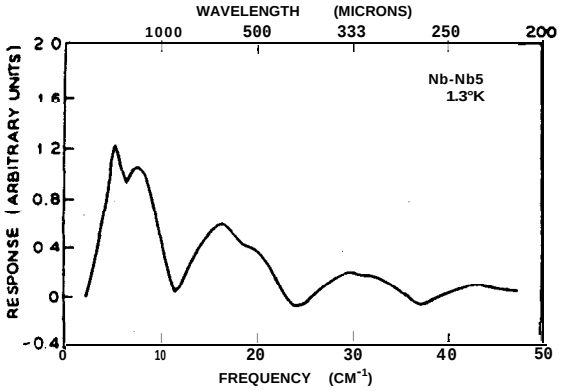


FIG. 8. Spectral response of a Nb-Nb point-contact Josephson junction.

pointed out that the amplitude of the ac Josephson current was frequency dependent and should increase with increasing frequency to a singularity, or peak, at the energy gap. They further showed that above the gap, the Josephson current amplitude would fall only gradually with increasing frequency and would still be appreciable at frequencies as high as three or four times the gap. Although different in detail, we suggest that our results on the strength of the response near the gap and the breadth of the frequency response of In-In detectors is evidence for the Riedel-Werthamer peak.

In Fig. 8 are shown data for a Nb-Nb junction. Despite some variability all Nb-Nb junctions yielded spectra much like Fig. 8. Niobium has a much higher transition temperature than In (9.1°K compared to 3.4°K) and consequently, a much larger energy gap (23 cm^{-1} compared to 7.2 cm^{-1} at our experimental temperatures). We would thus expect the response to extend to higher frequencies for Nb-Nb than for In-In junctions and it does, being still appreciable at frequencies beyond 30 cm^{-1} or wavelengths smaller than 300 μ. The form of the response curve, however, is very different from that of In or from the Riedel-Werthamer calculations. In fact, instead of a peak in the response at the gap there are minima at the gap and

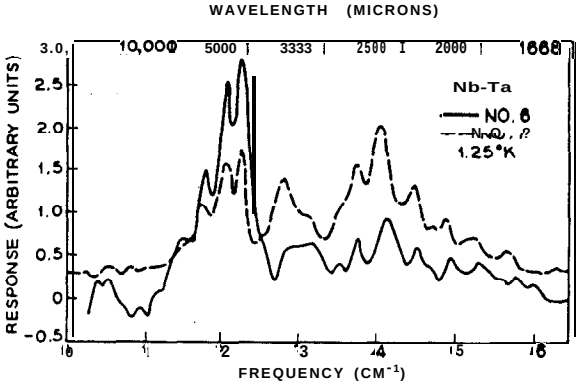


FIG. 9. Spectral response of a Nb-Ta point-contact Josephson junction.

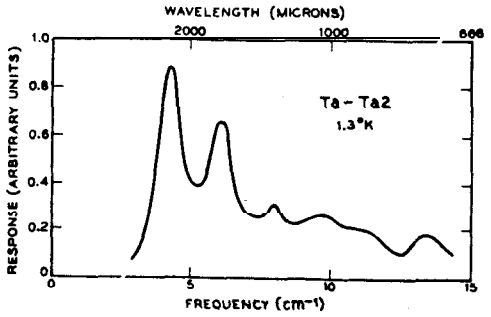


FIG. 10. Spectral response of a Ta-Ta Point-contact Josephson junction.

at $\frac{1}{2}$ the gap, $\frac{1}{4}$ the gap, and $\frac{3}{2}$ the gap as well. However, regardless of the shape of the response curve, it is true that Nb-Nb junctions also respond out to about twice the gap as did the indium junctions.

To supplement these ir spectral response results, we were able, with the collaboration of M. A. Pollack, to carry out some preliminary experiments using an acetonitrile gas laser as a monochromatic source at 32.2 cm^{-1}. We found two constant-voltage steps in the V-I characteristic of a Nb-Nb junction at voltages corresponding to the laser frequency and its second harmonic, but otherwise resembling the steps shown in Fig. 2 for 5 cm^{-1}. In effect, this checks one point on the spectral response curve. Note that in these laser experiments the monochromatic frequency applied was already well above the gap, yet two eonstant-voltage steps arising from the ac Josephson effect were seen. Using the same laser operating on the water-vapor transitions at 45.5 cm^{-1} and 84.9 cm^{-1}, we failed to see any constant-voltage steps.

Spectral response curves for junctions formed from a number of other superconductors were also obtained. In each case the curves were different (both from the data shown up to now and from the Riedel-Werthamer theory) .

Figure 9 shows the spectral response curve for a Nb-Ta junction. The energy gap for Ta is 10.7 cm^{-1} so the response is at frequencies well below both the energy gaps involved. Note, however, the prominent

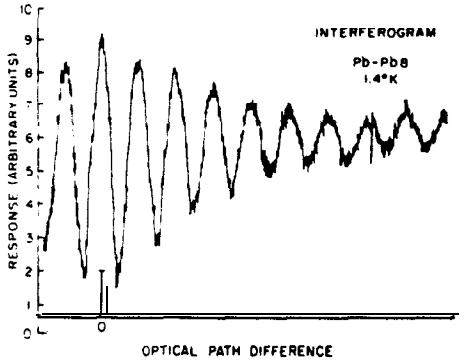


FIG. 11. Portion of an interferogram for a Pb-Pb point-contact Josephson junction.

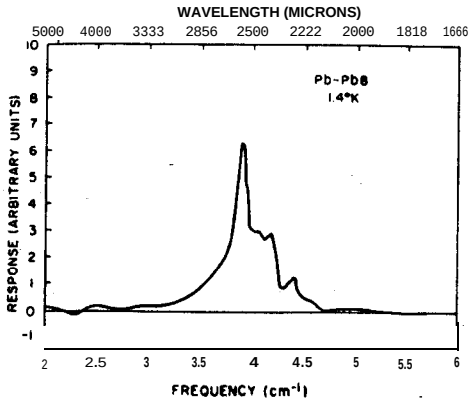


FIG. 12. Spectral response of the Pb-Pb point-contact Josephson junction of Fig. 11.

structure in the solid curve at about 2 cm^{-1} and the presence of similar structure at just twice the frequency. The dashed curve was taken for the same junction by applying a small magnetic field to counteract the effect of some trapped flux and to increase the Josephson current. Again, the harmonically related structure is present. In fact, the second harmonic has been emphasized relative to the fundamental.

Figure 10 shows data for a Ta-Ta junction. Again the response is largely at frequencies below the energy gap and there is some indication of periodicity in the response.

Spectra were also taken for Pb-Pb junctions and invariably they revealed very strong, very sharp response curves at a frequency far below the Pb gap of 21.5 cm^{-1}. Figure 11 shows a portion only of an interferogram for a Pb-Pb junction. That there is a dominant frequency in the spectrum and that the Q is rather high is evident from the slow decrease in amplitude of succeeding interference peaks. The resulting spectrum is shown in Fig. 12.

There is a strong response at essentially one frequency which is well below the gap. Upon repeating the measurement with other Pb-Pb junctions, there seemed to be a strong tendericy always to get a dominant response at some low frequency. We have seen such peaks as low as 1.5 cm^{-1} and as high as 14 cm^{-1} though usually not as narrow as in Fig. 12.

The data for Pb are complicated by several factors. Because the Pb points are soft and not, covered with a tough oxide layer it was very difficult to obtain desirable V-I characteristics. All Pb-Pb junctions contained hysteresis regions in their V-I curves.[19] It is perhaps significant that spectra for In-In junctions that had hysteresis regions in their V-I curves were similar to the Pb-Pb spectra in that they also tended to yield a prominent peak and, generally, at some frequency well below the gap. Yet Nb-Nb junctions gave rise to spectra similar to that of Fig. 8 whether or not their V-I curves had a hysteresis region. Further complicating the situation is the fact that data for all Pb-Pb junctions save one, and for many of the In-In junctions were taken using a mounting geometry known to posses several strong cavity resonances. The one Pb-Pb junction for which data were taken in the conical mounting had its peak response at a higher frequency than and other Pb-Pb junction tested.

We believe that the narrow response peaks seen for Pb as well as for some In junctions, the harmonically related peaks in Nb-Ta junctions, and the suggestion of periodicity in the response of Ta-Ta and Nb-Nb junctions reflect the fact that a Josephson junction is an active device. It is well known that a Josephson junction will generate high-frequency radiation, especially if it is coupled to a resonant circuit, even if the Q of the resonance is low.[4–6,20] Such resonances certainly exist in all the mounting geometries employed in our experiments. Therefore the junctions may be generating some radiation of their own as well as detecting the incident radiation and thereby giving rise to complicated response curves. In an effort to test this hypothesis. experiments with pairs of junctions were carried out and are described in Sec. V.

Regardless of the ultimate explanation for the exact form of the response curves, these ir experiments demonstrate that Josephson currents exist at frequencies well beyond the superconducting energy gap. Also, considering the low power available from standard arc lamps, they indicate that Josephson junctions are sensitive detectors of millimeter and submillimeter radiation.

IV. DETECTOR SENSITMTY AND SPEED OF RESPONSE

We have measured the video sensitivity of a Nb-Nb junction detector at 2.5 cm^{-1} using microwave techniques. The detection system was identical to that used in the far-infrared experiments with the radiation in this case "chopped" by a ferrite gyrator to which a 90 Hz square-wave current was applied as a modulating signal. Calibrated attenuators accurate to a few percent were used to control the power level at the detector. The junction was mounted across the narrow dimension of RG98 waveguide (0.148 in. X 0.074 in.) which was left open at its termination about five wavelengths beyond the junction. The junction was immersed directly in liquid helium and its V-I curve was adjusted by means of a diderential screw mechanism as in the in experiments.

A number of measurements were made at various

[19] In taking data for junctions with a hysteresis region in their V-I curves, an ac current modulation at about 1000 Hz was superimposed on the junction dc bias so that the hysteresis region was traced out many times during each cycle of the 90 Hz chopper. The detector output thus consisted of an average voltage, related to the area of the hysteresis loop, which depended upon the amplitude of the maximum zero-voltage current.

[20] D. N. Langenberg, D. J. Scalapino, and B. N. Taylor, Proc. IEEE 54, 560 (1966).

signal-to-noise levels and then extrapolated. We found, with a time constant of 1 sec, a noise-equivalent power of 5×10^{-13} W, a value which already compares favorably with that of other helium temperature detectors even though the frequency of the measurement was well below that at which niobium had its maximum response.

Several points about this measurement deserve comment. The quoted figure does not represent any inherent detector sensitivity limit but rather the result of one particular esperimental configuration. As in our in measurements, the detector was not, in any sense, matched to the incident radiation. The most important source of noise came from the sensitivity of the junction characteristic to mechanical shock and vibration and to electrical interference from signals generated outside the immediate laboratory environment. Despite our efforts at shielding, occasional strong esternal signals "washed out" the V-I curve to such an extent that experimentation became impossible.

Most important, however, in achieving high detector sensitivity is care in adjusting the detector characteristic. Unfortunately, no universally applicable guide lines can be given to distinguish "sensitive" from "insensitive" characteristics. With experience, however, it is possible to make a reasonably reliable judgment about any given characteristic. Some of the features looked for are symmetry of the characteristic, indicative of the absence of trapped flux in the junction, zero-voltage currents of no more than a few milliamperes, with the attendant junction resistance of at least several ohms, and, most important of all, a sharp, sudden onset of voltage at a well-defined maximum zero-voltage current. These features largely concern the conversion of the small change in zero-voltage current to as large as possible a signal voltage across the detector. The basic detection mechanism for all junctions is the same : the response of zero-voltage Josephson current to incident radiation. Thus the device is *not* a thermal detector and could be used as a narrow-band or heterodyne detector with suitably designed apparatus.

In view of the Josephson-effect detection mechanism, the known ability[5,6] of point-contact Josephson junctions to generate radiation at frequencies of at least 10 GHz and their broad-band frequency response as detectors, the Josephson-effect detector ought to have a very small response time. We carried out several experiments using Nb-Nb junctions to set a reasonable lower limit to the detector speed of response. These experiments were, again, at 2.5 cm^{-1} and used microwave components and techniques. We found that the detector easily responded to amplitude modulation of the microwave signal at a 1 MHz modulation frequency. We were also able to apply a voltage pulse of about 100 nsec duration to the reflector electrode of the klystron. This produced a pulse of 2.5 cm^{-1} radiation with a risetime of 10 nsec, as observed with a conventional room-temperature crystal detector. When this pulse of radiation was applied repetitively to a Nb-Nb junction and the junction output voltage was observed on a sampling oscilloscope, it was found that the Josephson-effect detector reproduced the 10 nsec risetime of the pulse. We conclude that the response time of the detector is less than 10 nsec.

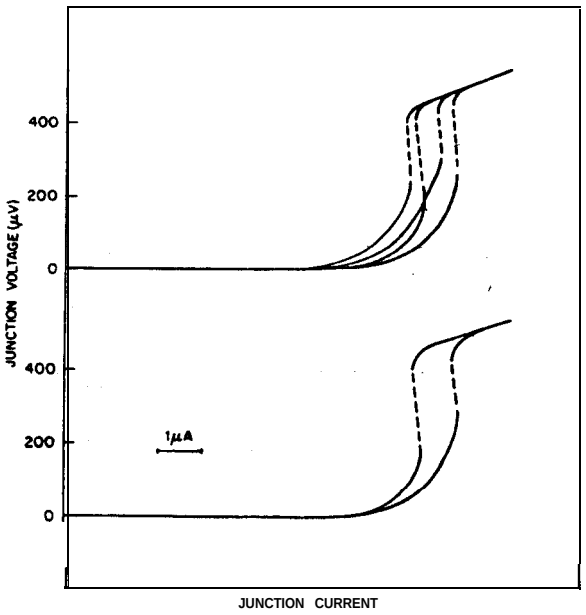


FIG. 13. Result of experiment designed to show that point-contact junctions are emitters as well as detectors of radiation. The lower trace is a portion of the *V-I* curve of the detector junction. The upper trace is the same portion but now with nearby emitter junction being switched on and off. Radiation from emitter decreases the detector zero-voltage current.

V. TWO-JUNCTION EXPERIMENTS

To test whether the point-contact Josephson junctions could have been emitting radiation and thus complicating their response curves as detectors, a number of experiments were carried out involving two independent and physically separated junctions. In each case separate leads, separate biasing, amplifying and display equipment, and separate differential screw adjustment mechanisms were used.

In one series of experiments, the two distinct junctions, separated by about $\frac{1}{2}$ in., were located in the same metallic bos, which was entirely closed except for the small insulated holes through which the wires forming the junctions were introduced. One of the junctions was treated as an emitter by using square-wave bias that switched the junction voltage alternately between zero and a finite voltage whose value was adjustable. Thus the emitter junction was turned on and off at a convenient audio frequency, while the detector junction voltage was monitored either on an oscilloscope or by a lock-in amplifier synchronized to the emitter square-wave frequency.

Results for a pair of Nb-Nb junctions are shown in Figs. 13 and 14. The lower trace in Fig. 13 is a portion of

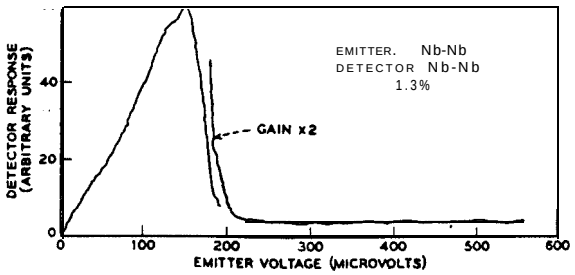


FIG. **14. Dependence of detector junction response on emitter junction bias voltage.**

the detector V-I curve with the emitter inoperative, i.e., with no voltage applied to the emitter junction. This trace shows appreciable hysteresis. The upper trace shows the same portion of the detector V-I curve with the emitter junction now being turned on and off. In this case the curves with the emitter off and on are superimposed. With the emitter off, the curve is identical with that in the lower trace; the shift produced with the emitter on is. readily apparent. Figure 14 shows the dependence of the detector response on the emitter bias voltage. The data were obtained by slowly sweeping the amplitude of the emitter square-wave bias and plotting the output of the detector against the dc voltage obtained by rectifying and amplifying the emitter voltage. The data of Fig. 14 and similar data plotted for other pairs of junctions follow a curve which is identical, as far as our limited accuracy permits us to make the comparison, with a curve of the emitter differential resistance dV/dI vs emitter voltage. Thus, the detector output depends on the emitter resistance and so accounts for the three major features of Fig. 14, namely, the finite slope at zero emitter voltage, the peak in response at a few tenths of a millivolt, and the drop to a constant response level at higher voltages.

The two-junction experiments were repeated in a configuration in which the junctions, separated by about 12 cm, were both mounted across the narrow dimension of a 4 mm waveguide. Again a result of the form of Fig. 14 was obtained. This result is not consistent with the idea of radiation being emitted by one junction at the Josephson frequency and detected by the other, since the waveguide cutoff at about 1.3 cm^{-1} should have been apparent in the curve. Using this configuration, we were able to interchange roles of emitter and detector and obtain results similar to Fig. 14 regardless of which junction was emitter or which was detector.

Again the two-junction experiments were repeated, now with each junction situated in a separate rig in its own Dewar system. A waveguide connection was introduced, at room temperature, between the two Dewars and their respective junctions. Again a result similar to Fig. 14 was obtained, although now only with great difficulty and with much reduced intensity. The effect of the emitter on the detector was completely unaffected when the waveguide connection between the two was removed.

We are led to conclude from these two-junction experiments that the communication link between emitter and detector, which was undoubtedly present, was inductive coupling via electrical leads of near-field radiation at a frequency most probably of a few tens of megacycles due either to a negative-resistance oscillation[7] or to radiation at subharmonics of the Josephson frequency.[6] Still left unexplained however is the dependence of the detector output on the emitter differential resistance.

In summary, then, these two-junction experiments, despite the questions they leave unanswered, do demonstrate that the junctions behave as active devices and suggest that the variability in form frqm superconductor-to-superconductor of the detector spectral response curves and the variability in attainable sensitivity from junction-to-junction may result from the complex nature of active junction behavior.

VI. CONCLUSION

Our experiments have demonstrated that the ac Josephson effect extends to frequencies at least twice the superconducting energy gap. We have made use of this effect in a far-infrared detector of high sensitivity and fast speed of response. Although far from any inherent limits, the performance figures already achieved for print-contact Josephson junction detectors are noteworthy. These figures include frequency response extending beyond 30 cm^{-1}, sensitivity better than 5×10^{-13} W noise-equivalent-power, and response time better than 10 nsec.

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